BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: KT-IX-23-1

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Code A

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and overwriting will not be awarded any marks.

Qi. Which of the following elements represent one of the rows of the matrix [9 5 9; 3 7 -1; 2 8 6]

(A) 9,3,2
(B) 5,7,8
(C) 9,5,9
(D) 9,7,6

Qii. Which of the following two matrices are equal?. E=[2+3 0+5; -6 4], F=[8-3 2+3; -6 4], G=[1+6 2+7; -6 4], H=[0+7 6+4; +6 4]

(A) G and H
(B) H and F
(C) F and E
(D) E and H

Qiii. Which of the following is a null matrix?

(A) [0 0; 0 0]
(B) [1 0; 0 1]
(C) [0 1; 1 0]
(D) [1 1; 1 1]

Qiv. If S = [12 9; -8 4], and T = [9 -7; -1 6], then S+T is equal to .....

(A) [3 2; 7 10]
(B) [21 2; -9 10]
(C) [21 -2; -7 10]
(D) [21 16; -9 -2]

Qv. The irrational number √5 can be expressed as .....

(A) 0.23606....
(B) 2.23606...
(C) 22.3606...
(D) 236.06...

Qvi. $\sqrt[3]{n^5}$ can be expressed in exponential form as ......

(A) $n^{7/5}$
(B) $n^{5/3}$
(C) $n^{2/3}$
(D) $n^{1/7}$

Qvii. If z = 8 + 3i then conjugate of z is .....

(A) -8 - 3i
(B) 8 + 3i
(C) 8 - 3i
(D) -3i

Qviii. 0.23987 can be written in scientific notation as .....

(A) 2.3987×10^{-1}
(B) 2.3987×10^{1}
(C) 23.987×10^{-3}
(D) 23.987×10^{1}

Qix. Which of the following is the base of natural logarithm ?.

(A) 5
(B) 10
(C) e
(D) g

Qx. Which of the following is a rational expression?.

(A) $(x^3+2x^2+5)/(7-y)$
(B) $\sqrt{(x^2+2x^2+3)}/(6-y)$
(C) $(x^3-2x^2-5)/(7-y)$
(D) $\sqrt{(x^3-2x^2-11)}/(11-y)$

Qxi. $(3m + n)^3$ = ...

(A) $9m^3 + n^3 +9mn(3m+n)$
(B) $9m^3 – n^3 -9mn(3m- n)$
(C) $9m^3 + n^3 – 9mn(3m+n)$
(D) $9m^3 – n^3 + 9mn(3m–n)$

Qxii. The factorization of $4y^2 – 12y + 9$ is .....

(A) (2y+3)(2y+3)
(B) (2y-3)(2y-3)
(C) (2y-3)(2y+3)
(D) (-2y+3)(2y-3)

Qxiii. If C and D are any two polynomials and their HCF and LCM are respectively H and L then C = ...

(A) L/D
(B) H/D
(C) (H×L)/D
(D) D/(H×L)

Qxiv. The solution of linear equation 2x + 5 = -4x + 3 is ...

(A) 4/3
(B) 1/3
(C) -1/3
(D) -4/3

Qxv. If X = {-1,0} and y = {5,7}, then which of the following is ordered pair of X × y?.

(A) {(-1,0),(5,0)}
(B) {(-1,5)}
(C) {(5,-1),(7,0)}
(D) {(7,0)}

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36

QQ.2. Attempt any NINE parts of the following. All parts carry equal marks.

i If C = [1 0; 5 1] and D = [2 6; 1 -7] determine whether CD = DC, or not?.

ii Simplify i. $(-4p^2 q)^5$ ii. $(-x^3 y^2)^3$

iii Find the volume of "p": log_√5 6 + log_√5 8 – log_√5 4 = log_√5 p

iv Evaluate the following when b = -2: $3b^3 – 4b^2 – 3$

v Find the value of 'pq' if p + q = 9, p – q = 7

vi Without performing division, find the remainder when $2w^2 – 5w + 3$ is divided by w – 3

vii Simplify: (c+d)/(3c+2d) + (c–d)/(3c+2d)

viii Solve: 8 + 3√c = 20

ix Solve and plot on number line: z – 5/7 ≤ (15+2z)/7

x Prove that the points p(3,1), Q(6,4) and R(9,7) are collinear.

xi Prove that if two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

xii Prove that any point on the right bisector of a line segment is equidistant from end points of the segment.

Section "C"

Maximum Marks: 24

Note: Note: Attempt any FOUR questions of the following. All questions carry equal marks.

QQ.3. Prove that if two opposite sides of a quadrilateral are congruent, then it is a parallelogram.

QQ.4. Prove that from a point outside a line, the perpendicular is the shortest distance from the point to the line.

QQ.5. Prove that if a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side.

QQ.6. Prove that in a right-angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

QQ.7. Prove that parallelograms on equal bases and having the same altitude are equal in area.

QQ.8. Construct ΔLMN, for mLM = 5.4cm, m∠L = 75° and m∠M = 45°.